

Anshul Joshi

Software Engineer | GenAI & Conversational Systems | Applied ML Researcher
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Education

University of Illinois Urbana-Champaign

Master of Science in Computer Science

Aug 2026 – May 2028 (Expected)

Delhi Technological University

B.Tech in Computer Science • GPA: 9.02/10 • Department Rank 1, DTU Merit Scholar (2020–21)

Jul 2019 – May 2023

Experience

Software Engineer II

American Express – Global Interactive Voice Response Team

Apr 2025 – Present

- Architected a Multi-Modal Voice Response (MMVR) platform enabling synchronized voice and in-app visual experiences across internet and intranet channels, executing fintech functionalities (e.g., payments) with enterprise security controls; drove a **20% increase in self-service rate** while sustaining **100K+ calls** at **99.9% availability**.
- Designed and delivered an API orchestration and session-spanning framework supporting parallel voice and web ecosystems.

Software Engineer I

American Express – Global Interactive Voice Response Team

Jul 2023 – Apr 2025

- Engineered an LLM-based self-service voice bot integrating RASA and OpenAI models (Whisper TTS, Nuance STT), replacing static IVR scripts with context-aware prompting within Amex's regulatory environment.
- Fine-tuned intent and entity recognition using LLMs across the enterprise, increasing match rate by **29%** over the legacy ML baseline.
- Led Acquisition Promotions and card Freeze/Unfreeze microservices and global call-flow development supporting **5,000+ calls/month** at **99.9% availability**, achieving **150% higher call volume than projected**, **8K–10K annual call containment**, and **\$80K–\$100K yearly savings**.

Software Engineer Intern

American Express

May 2022 – Jul 2022

- Developed a real-time social-media interaction bot (Python, Tweepy) to monitor tweets mentioning Amex; applied multi-factor classification across 5+ signals (sentiment, virality, bot probability, prior engagement) to automate actions, analyzed **50,000+** customer and competitor tweets, and prototyped an agent context layer to improve pre-interaction visibility.

Technical Advisor & Business Development Analyst

Sereko – India's First Psychodermatology Brand

Dec 2022 – Feb 2023

- Conducted price benchmarking across 50+ competitor SKUs, finalized 20+ launch SKUs, and designed an SEO + LLM-based discovery strategy; evaluated 10+ marketplaces to support pre-launch 0→1 go-to-market setup.

Research Experience

Research Intern Fellow

Delhi Technological University

May 2021 – Aug 2021

- Researched *Fake News Detection Using Machine Learning and Metaheuristic Optimization Techniques*, benchmarking state-of-the-art optimization algorithms and developing a novel hybrid metaheuristic model.

Research Publications

- Credit Scoring using Swarm Intelligence for Feature Selection and PSO-Trained ANN Classification.** ViTE-CoN, IEEE, May 2023.
[DOI: 10.1109/ViTECoN58111.2023.10157909](https://doi.org/10.1109/ViTECoN58111.2023.10157909)
- Social Media Interactive Resource Planner for Students (SMIRP).** ICACITE, IEEE, Aug 2024.
[DOI: 10.1109/ICACITE60783.2024.10616807](https://doi.org/10.1109/ICACITE60783.2024.10616807)

Leadership & Social Impact

Co-Founder

PRJ Care Foundation – New Delhi, India

2017 – Present (registered 2022)

- Scaled the organization to impact **500+ underprivileged children**, driving **50+ health camps**, **1,000+ trees planted**, and **15+ recognitions**; launched a crowdfunding platform to diversify funding and sustain operations at scale.
- Spearheaded education initiatives enabling **100+ children** to achieve foundational literacy (English, Hindi, Math) and facilitated **25 first-time school admissions**; recruited and managed **100 college interns** while forging partnerships

with Manipal Hospital and DAV Public School.

Education Volunteer

Stand N Stride Foundation

- Imparted free basic education to underprivileged children up to high school under the “Pathshala” initiative.

Technical Skills

Languages: C++, Python, Java, JavaScript, MATLAB

Backend & Systems: Spring Boot, REST APIs, Microservices, Distributed Systems, Flask

Frontend: React, JavaScript, Redux

ML/AI: PyTorch, TensorFlow, Transformers, Diffusers, Scikit-learn, OpenCV, LLM fine-tuning, RASA

Cloud & DevOps: Git, Docker, Jenkins, GitHub Actions, XLR

Generative IDE: Claude Code, GitHub Copilot

Observability: Elasticsearch, Logstash, Kibana

Data: Pandas, NumPy, Matplotlib, SQL, Tableau

Optimization: Linear Programming, Optimization Models, Swarm Intelligence

Foundations: Data Structures & Algorithms, OOP, System Design (LLD/HLD), DBMS

Achievements

- **GRE:** 170/170 Quantitative (96+ percentile), 155/170 Verbal • **IELTS:** 8.5 band.
- **CAT 2025:** 99.3 percentile among 4 lakh+ applicants.
- **GrowthHack 2024:** Top 5% among 450+ teams in Amex’s largest company-wide hackathon.
- **CFA Level 1 scholar** • Finance Forward Quiz 2026 winner (Rank 1); demonstrated equity valuation and financial strategies for cost predictability and P&L analysis on trade orders.
- **Department Rank 1** & DTU Merit Scholar, Computer Science (2020–21).
- **JEE Mains:** 99+ percentile • Rank 76, Delhi Government Scholar (Junior Science Talent Search) • State Rank Top 10, Silverzone International Maths Olympiad.